

Research Article

Experimental Study on the Mechanical, Durability, and Microstructural Properties of Concrete With Partial Cement Replacement With Coffee Husk Ash and Rice Husk Ash

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The environmental impact of concrete production, mainly due to cement manufacturing, has prompted the search for sustainable alternatives, such as agricultural byproducts. Rice husk ash (RHA) and coffee husk ash (CHA) show promise as supplementary cementitious materials that can enhance concrete's strength and durability. However, RHA's fine particles reduce workability, and CHA's lower silica content limits its pozzolanic reactivity. While both have individual benefits, their combined effects on concrete properties remain underexplored. This study investigates the influence of RHA and CHA as partial cement replacements on the fresh and hardened properties of concrete. A C-25 mix was designed using the ACI 211.1 method, with RHA and CHA proportions ranging from 0% to 25% and 0% to 15%, respectively, in 5% increments. Workability was assessed using slump tests, and mechanical properties—including compressive, split tensile, and flexural strengths—as well as durability (measured through water absorption, strength, and weight loss under 5% sulfuric acid exposure) were evaluated. The results show that both RHA and CHA significantly affect workability, strength, and durability. RHA reduced workability more than CHA due to its finer particles and higher water absorption. The optimal mix of 10% RHA and 5% CHA improved compressive strength by 5.04% and 7.87% at 28 and 56 days, respectively, and enhanced flexural and split tensile strengths by 11.45% and 7.41%, respectively. Durability tests showed that the combination of 15% RHA and 5% CHA (R15C5) significantly reduced water absorption and improved sulfuric acid resistance, indicating a denser microstructure and greater environmental resilience. SEM analysis shows that adding RHA and CHA enhances concrete's microstructure. The R10C5 mix (10% RHA and 5% CHA) strikes a good balance of strength and refinement. While R15C5 (15% RHA and 5% CHA) improves durability, it shows similar strength and durability to R10C5 due to dilution and higher water demand, as confirmed by statistical analysis. These findings demonstrate that the synergistic use of RHA and CHA can optimize concrete performance, providing a sustainable alternative for enhancing both strength and durability. However, excessive use of RHA or CHA beyond the optimal proportions can compromise performance due to interference with hydration and workability.

Keywords: cement replacement; coffee husk ash (CHA); durability enhancement; mechanical properties; pozzolanic reactivity; rice husk ash (RHA); sustainable concrete

1. Introduction

Concrete remains a key material in construction, but its environmental impact—especially due to Portland cement production—is a serious concern [1, 2]. Cement manufacturing accounts for around 8% of global CO₂ emissions, significantly contributing to climate change. This

has increased pressure on the construction industry to adopt more sustainable materials (Tessema et al., 2022). As a result, alternative materials such as agricultural waste have garnered attention as sustainable, low-cost supplementary cementitious materials (SCMs) [3]. Notably, rice husk ash (RHA) [4] and coffee husk ash (CHA) [5] have received increased focus due to their environmental and economic

benefits, as well as their ability to enhance the properties of concrete.

RHA is a byproduct of the combustion of rice husks, which are agricultural residues obtained during rice milling [4]. The primary component of RHA is silicon dioxide [6, 7]. This high silica content is crucial for its pozzolanic properties, which contribute to enhanced compressive strength and durability in concrete [4]. Studies have demonstrated that incorporating RHA into concrete can improve compressive strength, reduce permeability, and enhance resistance to aggressive environmental conditions such as acid attacks [8, 9]. Additionally, RHA has been shown to improve the durability of concrete due to its ability to form calcium silicate hydrates (C-S-H) in the presence of calcium hydroxide [6]. However, the use of RHA as a cement replacement also presents challenges, particularly with regard to workability. The fine particle size and high surface area of RHA increase the water demand of the concrete mix, reducing the flowability and ease of mixing [10, 11]. Moreover, while RHA enhances the long-term strength of concrete, it may retard the early strength development [12]. Furthermore, Chindaprasirt et al. [13] showed that RHA improved sulfate resistance in concrete, enhancing resilience to environmental degradation.

Similar to RHA, CHA can partially replace cement to improve concrete's mechanical properties, including compressive, flexural, and split tensile strengths (STSs) [14, 15]. CHA offers better workability compared to RHA due to its lower specific surface area, which helps reduce the water demand of the mix. However, CHA has a lower silica content than RHA, limiting its pozzolanic reactivity [15, 16]. Additionally, the higher loss on ignition (LOI) in CHA means that the material contains a significant amount of unburned organic matter, which can further impact its binding properties in concrete [14]. As a result, while CHA can improve workability, it may not provide the same level of durability and strength enhancement as RHA. Kanthe et al. [17] found that incorporating both RHA and fly ash (FA) as part of a ternary mix improved both compressive strength and durability. This suggests that a similar approach—blending RHA and CHA—could enhance concrete properties.

The combined use of RHA and CHA in concrete mixtures has the potential to balance the advantages and limitations of each material. By blending the two ashes, it may be possible to optimize the properties of concrete, particularly in terms of workability, strength development, and durability. Swaminathan et al. [18] demonstrated that the use of RHA, alongside other pozzolanic materials, improved compressive strength and decreased permeability, suggesting that hybrid mixtures can offer superior mechanical and durability properties. Despite the promising potential of hybrid RHA and CHA mixtures, a significant research gap remains regarding the optimal blend ratio and its impact on both the fresh and hardened concrete properties. While studies have investigated the effects of each ash type individually [8, 14, 15], there is a lack of studies focused on the combined application of both materials. For instance, Onsong et al. [19] summarized the chemical compositions

of both RHA and CHA in past studies, revealing that RHA typically contains a higher percentage of SiO_2 than CHA. This difference in silica content is crucial for their pozzolanic activity, as higher silica content generally leads to better pozzolanic reactivity and stronger cementitious properties [16]. In terms of workability, CHA's higher organic content and lower specific surface area make it more effective in improving the flowability of concrete compared to RHA, which tends to increase the water demand of the mix [14]. Thus, the hybrid mixture of RHA and CHA could offer a balanced approach, providing both enhanced strength and improved workability. Given these considerations, further research is warranted to determine the optimal blend ratio and evaluate the combined effects of RHA and CHA on concrete performance. Accordingly, the present study aims to examine the impact of partially replacing cement with varying proportions of a hybrid RHA-CHA mixture on the fresh properties (slump test), mechanical properties (compressive strength, flexural strength, and STS), durability properties (water absorption and sulfuric acid resistance), and microstructural characteristics (scanning electron microscope [SEM] analysis) of concrete. By addressing these aspects, the study aims to contribute valuable insights into the combined use of RHA and CHA in concrete.

2. Materials and Methods

2.1. Materials. The materials used in this study included water, coarse aggregate, fine aggregate, cement, RHA, and CHA. Potable tap water, sourced from the Central Laboratory of the Ethiopian Construction Works Corporation, was used to achieve the desired workability. Coarse and fine aggregates, selected based on ASTM C33, were obtained from local suppliers, to ensure concrete strength and stability. Washed river sand, sourced from East Shewa, and well-graded coarse aggregate with a nominal size of 19 mm were chosen for the mixture. Ordinary Portland Cement (OPC) with a grade of 42.5, produced by the Dangote Cement Factory, was used as the control binder. RHA was collected from Buno Bedele, where it was burned in a controlled environment at 500°C – 600°C for 6–7 h, and then sieved through a 75-micron sieve to match the fineness of cement. Similarly, CHA was obtained from the same region, burned to a black-gray color, cooled, and sieved to the same fineness. All materials were carefully prepared to ensure consistency throughout the experimental process.

2.2. Methods

2.2.1. Material Characterizations. The engineering properties of the materials used in this study were tested according to various standardized methods to ensure their suitability for concrete production. For the coarse aggregate, sieve analysis was conducted in accordance with ASTM C136 to determine the particle size distribution. The results of the sieve analysis were within the grading limits specified by ASTM C33, confirming the quality of the coarse aggregate for concrete. Additionally, specific gravity and absorption tests were performed following ASTM C127-01 and ASTM

C128-01, with the bulk specific gravity measured at 2.71 and the water absorption at 1.23%, which are consistent with typical standards for coarse aggregates. The bulk density of the aggregate, measured according to ASTM C29, was found to be 1.54 g/cm^3 , which is within the acceptable range for concrete. Furthermore, moisture content, determined using ASTM C566, showed a moisture content of 0.5%, in line with industry standards for concrete aggregates.

For the fine aggregate, the silt content was determined following ASTM C117, and the result was 2.8%, which is below the maximum allowable limit of 3% as per ASTM C33, ensuring that the fine aggregate is suitable for concrete production. The fineness modulus, calculated based on ASTM C136, was found to be 2.9, which is within the permissible range of 2.3–3.1 for concrete sand. The specific gravity of the fine aggregate was tested following ASTM C128, and the result was 2.67, which is in accordance with ASTM C128 for fine aggregates. Absorption was found to be 1.45%, based on ASTM C128, which is typical for fine aggregates used in concrete. The fineness of the cement was tested according to ASTM C204, and the result was $2550\text{ cm}^2/\text{g}$, meeting the specifications required for Portland cement. In comparison, CHA had a fineness of $11,680\text{ cm}^2/\text{g}$, while RHA exhibited a much higher fineness of $233,000\text{ cm}^2/\text{g}$. According to Blesson and Rao (2024), the performance of alkali-activated ternary binder paste and mortar, consisting of CHA, RHA, and GGBS, showed conformance to the variations in specific surface area and fineness as described above. The specific gravities of RHA and CHA were measured at 2.12 and 2.45, respectively, which confirms their potential as viable partial replacements for cement in concrete production. The RHA and CHA were analyzed for their chemical compositions and specific gravities at the Ethiopian Mineralogy and Geotechnical Laboratory. Table 1 shows the oxide composition of RHA and CHA that were tested during the study. RHA and CHA must meet ASTM C618 requirements to be used as a pozzolan in concrete production. The chemical composition of RHA and CHA should have a combined proportion of silicon dioxide (SiO_2), aluminum oxide (Al_2O_3), and iron oxide (Fe_2O_3) of at least 70%. The LOI should not exceed 10%, as per ASTM standards. Class F pozzolan, which contains up to 12% LOI, is commonly used. The available alkalis, specifically Na_2O , should not exceed 1.5% according to ASTM C618. The MgO percentage in cement, originating from magnesia compounds in raw materials, typically ranges from 0.1% to 4%, with 5% being the maximum to control expansion in hardened concrete. The laboratory report indicated that the MgO content was within the acceptable limit. The LOI of both ashes revealed a result of less than 12%, which is lower than the maximum limit of 12% set by ASTM. For the cement, the consistency test was conducted using the Vicat apparatus to determine the standard consistency, and the result was 28%, which is within the normal range for Portland cement hydration.

2.2.2. Mix Design and Proportioning. To ensure the desired properties of concrete, several steps were undertaken to determine the appropriate mix proportions. The target compressive strength for the control mix was set at 25 MPa.

Following the ACI 211.1 method, a mix design was created for C-25 concrete, aiming for a target mean compressive strength of 33.5 MPa. The selected mix proportion was 1: 2.337:2.678 (cement: fine aggregate: coarse aggregate), with a water-to-cement ratio (w/c) of 0.491, making it suitable for general reinforced or structural concrete. In this study, the weight-batching method was used for mix proportioning, replacing OPC with varying percentages of CHA and RHA, as it provides greater accuracy than volume batching for materials with voids. This mix design is crucial as the success of partial cement replacements with materials such as RHA and CHA largely depends on achieving a balanced mix that meets the desired mechanical and durability properties. In a related study by Bixapathi and Saravanan [20], the effects of replacing OPC with RHA in M20-grade concrete were explored. They tested concrete mixes with varying proportions of RHA (0%, 5%, 10%, 15%, 20%, and 25%) and assessed their compressive strength, splitting tensile strength, flexural strength, and durability. Their findings showed that 15% RHA replacement resulted in the highest strength values, but any further increase in RHA content beyond 15% caused a decline in both strength and durability. This reduction was attributed to an overreplacement of cement, which reduces the availability of the primary binder and results in a less dense microstructure. Similarly, research by Demissew et al. [15] and Adelfio et al. [21] has highlighted that the partial replacement of OPC with CHA can also enhance the compressive strength of concrete, particularly with up to 10% CHA content. The compressive strength at 7, 14, and 28 days for the 10% CHA mixture exceeded the 25 MPa design strength of conventional concrete, indicating CHA's positive contribution to concrete performance [16, 22]. This reinforces the idea that the addition of SCMs such as RHA and CHA can improve both mechanical and durability properties, depending on the correct mix proportion. To explore the combined effects of these materials, this study developed a hybrid mix using different combinations of RHA and CHA. The mix ingredients per cubic meter (in kg) are listed in Table 2, while the sample sizes for the tests are shown in Table 3.

As presented in Table 2, the mix matrix ranged from 0% to 25% RHA and 0% to 15% CHA at 5% intervals, with each combination tested for compressive strength, STS, flexural strength, water absorption, and sulfuric acid resistance in terms of strength and weight loss. The results from these tests provided valuable insights into how these materials interact when used together, contributing to the development of an optimized concrete mix that improves performance in terms of fresh, mechanical, and durability properties.

2.2.3. Testing Procedure. For the mix preparation, the necessary ingredients—OPC, RHA, CHA, coarse aggregate, fine aggregate, and water—were carefully proportioned. A variety of hybrid RHA and CHA combinations were mixed to ensure uniform distribution. The entire mixing process followed the guidelines outlined in ASTM C192, ensuring proper hydration and a homogeneous mixture. The first key

TABLE 1: The oxide compositions of RHA and CHA.

Oxide	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	MgO	Na ₂ O	K ₂ O	MnO	P ₂ O ₅	TiO ₂	H ₂ O	LOI	SiO ₂ + Al ₂ O ₃ + Fe ₂ O ₃
RHA (%)	80.72	1.15	0.6	1.38	1.65	1	2	0.08	1.65	0.04	3.02	5.5	82.47
CHA (%)	59.85	7.4	3.5	6	1.4	1.05	4.5	0.05	2.4	0.06	3.6	9	70.75

TABLE 2: The various combinations of RHA and CHA mix and associated ingredients per cubic meter (kg).

Mix code	RHA, CHA, and cement (%)			Ingredients per cubic meter (kg)					
	RHA (%)	CHA (%)	Cement (%)	RHA	CHA	Cement	FA	CA	Water
C1 (0, 0)	0	0	100	0.00	0.00	376.70	880.53	1009.00	198.19
C2 (5, 0)	5	0	95	18.59	0.00	353.15	868.93	995.71	195.58
C3 (10, 0)	10	0	90	36.69	0.00	330.21	857.63	982.76	193.04
C4 (15, 0)	15	0	85	54.33	0.00	307.87	846.62	970.15	190.56
C5 (20, 0)	20	0	80	71.52	0.00	286.08	835.89	957.85	188.14
C6 (25, 0)	25	0	75	88.28	0.00	264.85	825.43	945.86	185.79
C7 (0, 5)	0	5	95	0.00	18.32	348.05	856.38	981.33	192.75
C8 (5, 5)	5	5	90	18.08	18.08	325.51	845.40	968.75	190.28
C9 (10, 5)	10	5	85	35.71	17.85	303.53	834.70	956.49	187.88
C10 (15, 5)	15	5	80	52.89	17.63	282.11	824.27	944.54	185.53
C11 (20, 5)	20	5	75	69.66	17.41	261.21	814.10	932.88	183.24
C12 (25, 5)	25	5	70	86.01	17.20	240.82	804.17	921.50	181.00
C13 (0, 10)	0	10	90	0.00	35.66	320.93	833.52	955.13	187.61
C14 (5, 10)	5	10	85	17.61	35.21	299.32	823.12	943.21	185.27
C15 (10, 10)	10	10	80	34.78	34.78	278.24	812.97	931.59	182.98
C16 (15, 10)	15	10	75	51.53	34.36	257.67	803.07	920.24	180.76
C17 (20, 10)	20	10	70	67.89	33.94	237.60	793.41	909.17	178.58
C18 (25, 10)	25	10	65	83.85	33.54	218.01	783.98	898.37	176.46
C19 (0, 15)	0	15	85	0.00	52.10	295.22	811.85	930.30	182.73
C20 (5, 15)	5	15	80	17.15	51.46	274.48	801.98	918.99	180.51
C21 (10, 15)	10	15	75	33.90	50.85	254.23	792.34	907.95	178.34
C22 (15, 15)	15	15	70	50.24	50.24	234.46	782.94	897.17	176.22
C23 (20, 15)	20	15	65	66.20	49.65	215.16	773.75	886.64	174.16
C24 (25, 15)	25	15	60	81.80	49.08	196.31	764.78	876.36	172.14

test was the slump test, which evaluated the workability of the concrete. By incorporating different percentages of hybrid RHA and CHA, the experiment aimed to determine how these materials influenced the flowability and ease of placement. The test was performed according to ASTM C143/C143M-20 standards, helping gauge the effect of the added materials on the concrete's workability.

Once the concrete was mixed, it was poured into molds for various tests, such as cube molds for compressive strength, cylinder molds for STS, and beam molds for flexural strength. The workability was checked before pouring the mixture into the molds, using a slump cone and compacting the concrete through tamping to eliminate voids. Afterward, the concrete specimens were cured in accordance with ASTM C192/C192M-23 standards to promote hydration and strength development. Compressive strength tests, performed per ASTM C39/C39M-23, evaluated the concrete's ability to withstand compressive forces at 7, 28, and 56 days. The STS tests followed ASTM C496, providing data on the tensile properties of the concrete at 28 days. Both tests were carried out to offer valuable insights into the material's performance at these stages. In addition, flexural strength was assessed using the ASTM C78/C78M-22 standard at 28 days, contributing to the

understanding of the concrete's bending properties. Water absorption was tested on specimens cured for 7, 28, and 56 days according to ASTM C642. The acid resistance of C-25 concrete with RHA and CHA was tested by first curing 100-mm cubical specimens in water for 7 days. After this curing period, the specimens were immersed in a 5% sulfuric acid solution for an exposure period of 28 days. The solution, prepared using 98% concentrated sulfuric acid, was renewed regularly to maintain a pH below 7. The testing followed the modified ASTM C 267 (1997) method. Changes in weight and compressive strength were recorded at 28 days of acid exposure and compared with the strength of water-cured specimens.

A one-way ANOVA was conducted to compare the control mix (R0C0) with concrete mixes incorporating varying percentages of CHA and RHA, evaluating their individual and combined effects on compressive strength (7, 28, and 56 days), STS (28 days), and flexural strength (28 days). SEM analysis was performed to examine the microstructure and hydration characteristics of R0C0, R10C5, and R15C5 mixes using 28-day specimens. Additionally, an independent *t* test was used to assess the significance of strength differences between the R10C5 and R15C5 mixes.

TABLE 3: The various combinations of RHA and CHA mix and the number of samples for various tests.

Mix code	Cement replacement			Compressive			Split tensile	Flexural	Water absorption			Sulfuric
	RHA (%)	CHA (%)	Cement (%)	7 th	28 th	56 th	28 th	28 th	7 th	28 th	56 th	28 th
C1 (0, 0)	0	0	100	3	3	3	3	3	3	3	3	6
C2 (5, 0)	5	0	95	3	3	3	3	3	3	3	3	6
C3 (10, 0)	10	0	90	3	3	3	3	3	3	3	3	6
C4 (15, 0)	15	0	85	3	3	3	3	3	3	3	3	6
C5 (20, 0)	20	0	80	3	3	3	3	3	3	3	3	6
C6 (25, 0)	25	0	75	3	3	3	3	3	3	3	3	6
C7 (0, 5)	0	5	95	3	3	3	3	3	3	3	3	6
C8 (5, 5)	5	5	90	3	3	3	3	3	3	3	3	6
C9 (10, 5)	10	5	85	3	3	3	3	3	3	3	3	6
C10 (15, 5)	15	5	80	3	3	3	3	3	3	3	3	6
C11 (20, 5)	20	5	75	3	3	3	3	3	3	3	3	6
C12 (25, 5)	25	5	70	3	3	3	3	3	3	3	3	6
C13 (0, 10)	0	10	90	3	3	3	3	3	3	3	3	6
C14 (5, 10)	5	10	85	3	3	3	3	3	3	3	3	6
C15 (10, 10)	10	10	80	3	3	3	3	3	3	3	3	6
C16 (15, 10)	15	10	75	3	3	3	3	3	3	3	3	6
C17 (20, 10)	20	10	70	3	3	3	3	3	3	3	3	6
C18 (25, 10)	25	10	65	3	3	3	3	3	3	3	3	6
C19 (0, 15)	0	15	85	3	3	3	3	3	3	3	3	6
C20 (5, 15)	5	15	80	3	3	3	3	3	3	3	3	6
C21 (10, 15)	10	15	75	3	3	3	3	3	3	3	3	6
C22 (15, 15)	15	15	70	3	3	3	3	3	3	3	3	6
C23 (20, 15)	20	15	65	3	3	3	3	3	3	3	3	6
C24 (25, 15)	25	15	60	3	3	3	3	3	3	3	3	6

3. Results and Discussion

3.1. Influence of RHA and CHA on Workability. The effect of partial replacement of cement with RHA and CHA on the workability of different concrete mixes was thoroughly examined. The findings from the slump test results, as shown in Figure 1, indicate that as the percentage of RHA increases, the slump value decreases, which suggests that RHA significantly reduces the workability of the concrete. This reduction in slump is due to RHA's fine particles, which absorb more water and create a stiffer mix. Past studies support this observation, with Hakeem et al. [10] noting a decrease in slump flow as RHA was incorporated into concrete mixtures, reflecting reduced workability. Similarly, Raman et al. [23] explained that RHA's pozzolanic nature increases the water demand by reacting with available water, thus consuming part of water and reducing fluidity. Additionally, the fine particles and high surface area of RHA contribute to its ability to absorb water, further decreasing available water for lubrication, as reported by Obilade [11]. The irregular shape and angularity of RHA particles also increase friction, impeding flow and further reducing workability. Swaminathan et al. [18] also reported that increasing the proportions of MK and RHA in high-performance concrete (HPC) blends significantly reduced workability, as evidenced by lower slump, compaction factor, and Vee-Bee time, necessitating the use of superplasticizers to maintain consistency.

In contrast, CHA has a lesser impact on workability, as its coarser particles absorb less water. However, as the CHA content increases, a slight reduction in slump is observed. Asfaw et al. [14] observed a decrease in workability with increasing CHA, attributed to the higher specific surface area and water demand of CHA compared to cement. CHA's fine particles absorb more water, reducing available water for lubrication and plasticity. Tarekegn et al. [24] similarly noted decreased workability and density with higher hybrid ash of CHA and sugarcane bagasse ash, suggesting a higher water demand for adequate mixing and placement. When RHA and CHA are used in combination, the slump values are higher compared to using RHA alone at the same cement replacement percentage.

3.2. Effect of RHA and CHA on Compressive Strength. This study on the incorporation of CHA and RHA in concrete mixtures shows notable findings in compressive strength, as shown in Figure 2, with data at 7, 28, and 56 days and a standard deviation range of 0.04–1.35. The highest compressive strengths achieved were 36.25 MPa and 37.98 MPa at 28 and 56 days, respectively, representing increments of 5.04% and 7.87% compared to the control concrete. These improvements were observed with 10% RHA and 5% CHA (Table 4), which underscores the positive influence of moderate dosages of these materials on concrete strength. This aligns with previous studies, such as Demissew

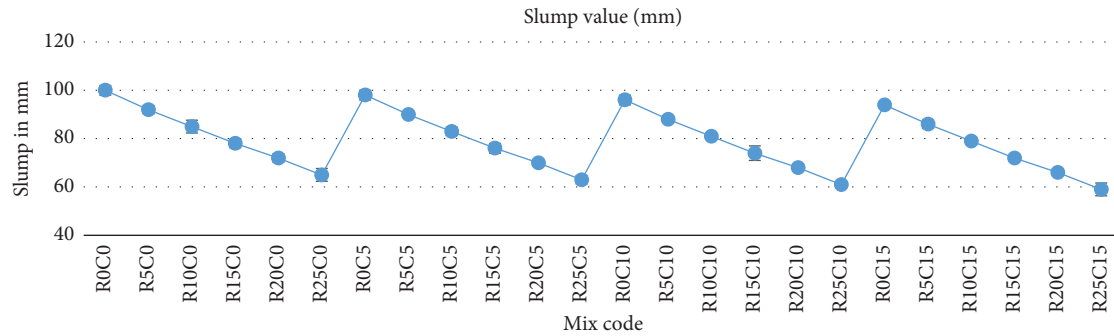


FIGURE 1: Influence of RHA and CHA on concrete slump.

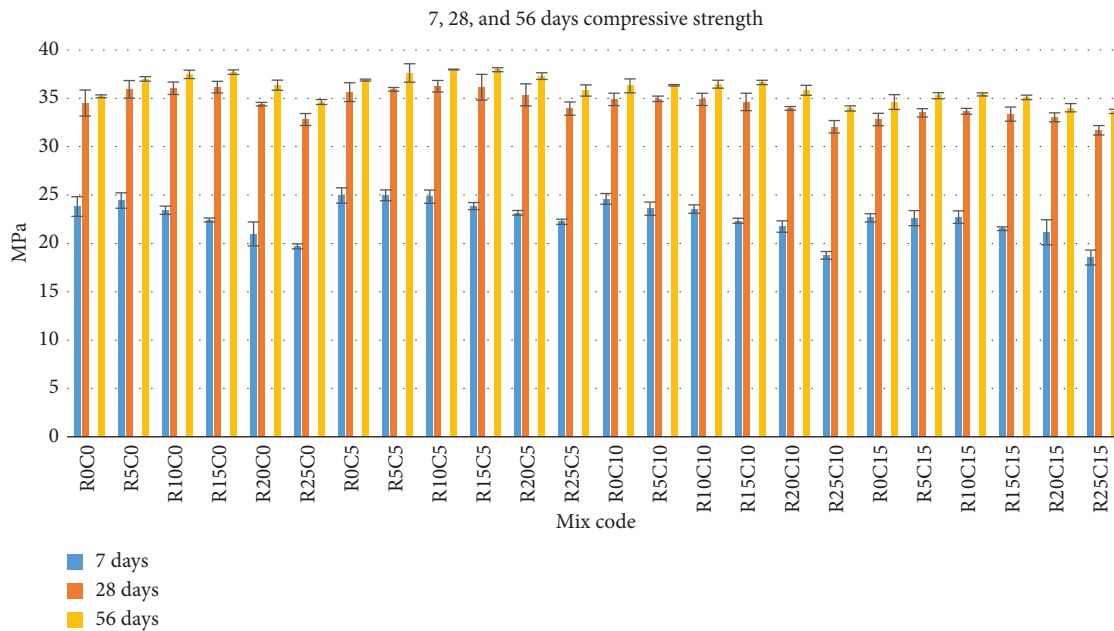


FIGURE 2: RHA and CHA effect on 7-, 28-, and 56-day compressive strength concrete.

et al. [15], who found that 10% CHA enhances compressive strength through its pozzolanic effects. Similarly, Zareei et al. [8] and Fapohunda et al. [4] observed that RHA's high silica content improves the concrete's strength by filling voids and refining its microstructure, leading to better interparticle bonding. The combination of 5% CHA and 10% RHA in this study optimized the pozzolanic effects of RHA while preventing the excessive carbon in CHA from disrupting the cement hydration process, in line with findings from Zareei et al. [8], who noted the densifying and strengthening effects of RHA's pozzolanic reaction. Khan and Haq [25] found that although RHA slightly reduces early-age compressive strength up to 28 days, it significantly enhances long-term strength, especially when combined with metakaolin (MK) or silica fume (SF). A similar trend was observed by Swaminathan et al. [18], where HPC blended with both RHA and MK showed superior compressive strength at all curing ages, with the optimum replacement level found to be 7.5% MK

and 12.5% RHA. Kanthe et al. [17] further supported this by showing that ternary blends with RHA and FA outperform binary and control mixes due to better particle packing and pozzolanic activity.

However, when higher dosages of CHA and RHA were used, the results showed a decline in compressive strength. Specifically, at 25% RHA and 15% CHA, the compressive strength dropped to 31.70 MPa and 33.65 MPa at 28 and 56 days, representing reductions of 8.15% and 4.43%, respectively. This decrease aligns with the findings of Demissew et al. [15], who noted that higher levels of CHA can impair concrete strength, likely due to excess carbon disrupting cement hydration. Similarly, Reta and Mahto [16] observed a comparable reduction in strength when CHA dosages exceeded 15%, further supporting the view that excess CHA interferes with hydration. Lo et al. [9] also emphasized that excessive pozzolanic materials can increase water demand and disrupt cement hydration, which can lead to weaker concrete.

TABLE 4: Change in compressive, split, and flexural strength in MPa from control concrete in respective ages (%).

Mix code	Compressive strength						Flexural strength						Split tensile strength					
	7 days			28 days			56 days			28 days			28 days			28 days		
	Average	Change (%)	p value	Average	Change (%)	p value	Average	Change (%)	p value	Average	Change (%)	p value	Average	Change (%)	p value	Average	Change (%)	p value
R0C0	23.81	0.00	—	34.51	0.00	—	35.21	0.00	—	4.28	0.00	—	4.05	0.00	—	4.05	0.00	—
R5C0	24.43	2.61	0.25	35.93	4.11	0.05	37.01	5.10	< 0.001	4.39	2.57	0.44	4.15	2.47	0.01	4.15	2.47	0.01
R10C0	23.43	-1.60	0.2	36.04	4.44	0.01	37.48	6.46	< 0.001	4.59	7.24	0.13	4.25	4.94	< 0.001	4.25	4.94	< 0.001
R15C0	22.42	-5.85	< 0.001	36.16	4.77	0.01	37.71	7.11	< 0.001	4.47	4.44	0.48	4.18	3.21	0.03	4.18	3.21	0.03
R20C0	20.98	-11.87	0.02	34.40	-0.32	0.35	36.36	3.27	0.02	4.19	-2.10	0.60	4	-1.23	0.08	4	-1.23	0.08
R25C0	19.69	-17.31	< 0.001	32.81	-4.92	0.01	34.62	-1.68	0.03	4.01	-6.31	0.11	3.85	-4.94	0.02	3.85	-4.94	0.02
R0C5	24.95	4.79	0.07	35.64	3.29	0.12	36.89	4.78	< 0.001	4.29	0.23	0.94	4.1	1.23	0.08	4.1	1.23	0.08
R5C5	24.97	4.87	0.02	35.93	4.11	0.00	37.62	6.84	0.01	4.6	7.48	0.23	4.2	3.70	0.00	4.2	3.70	0.00
R10C5	24.83	4.29	0.06	36.25	5.04	0.01	37.98	7.87	< 0.001	4.77	11.45	0.02	4.35	7.41	0.04	4.35	7.41	0.04
R15C5	23.86	0.21	0.82	36.15	4.76	0.10	37.96	7.81	< 0.001	4.74	10.75	0.07	4.33	6.91	< 0.001	4.33	6.91	< 0.001
R20C5	23.16	-2.73	0.01	35.36	2.46	0.26	37.30	5.95	< 0.001	4.39	2.57	0.52	4.05	0.00	1.00	4.05	0.00	1.00
R25C5	22.23	-6.62	< 0.001	33.94	-1.64	0.22	35.81	1.71	0.16	4.39	2.57	0.68	4	-1.23	0.90	4	-1.23	0.90
R0C10	24.60	3.32	0.07	34.89	1.11	0.36	36.29	3.07	0.06	4.31	0.70	0.92	4.12	1.73	0.05	4.12	1.73	0.05
R5C10	23.59	-0.92	0.61	34.95	1.28	0.05	36.35	3.23	< 0.001	4.52	5.61	0.15	4.25	4.94	< 0.001	4.25	4.94	< 0.001
R10C10	23.55	-1.08	0.36	34.89	1.11	0.36	36.46	3.56	0.01	4.73	10.51	0.03	4.35	7.41	< 0.001	4.35	7.41	< 0.001
R15C10	22.34	-6.19	< 0.001	34.63	0.35	0.83	36.64	4.06	< 0.001	4.61	7.71	0.58	4.28	5.68	0.14	4.28	5.68	0.14
R20C10	21.74	-8.71	< 0.001	33.96	-1.59	0.01	35.83	1.76	0.12	4.31	0.70	0.88	4.05	0.00	1.00	4.05	0.00	1.00
R25C10	18.76	-21.22	< 0.001	32.06	-7.09	< 0.001	33.95	-3.57	< 0.001	4.21	-1.64	0.62	3.95	-2.47	0.02	3.95	-2.47	0.02
R0C15	22.64	-4.91	0.01	32.81	-4.92	0.01	34.62	-1.68	0.26	4.01	-6.31	0.46	4	-1.23	0.06	4	-1.23	0.06
R5C15	22.61	-5.04	0.06	33.50	-2.94	0.02	35.27	0.18	0.78	4.13	-3.50	0.30	4.05	0.00	1.00	4.05	0.00	1.00
R10C15	22.72	-4.58	0.04	33.66	-2.47	0.01	35.41	0.56	0.19	4.07	-4.91	0.52	4	-1.23	0.20	4	-1.23	0.20
R15C15	21.52	-9.61	< 0.001	33.37	-3.31	0.05	35.10	-0.30	0.53	3.88	-9.35	0.05	3.75	-7.41	< 0.001	3.75	-7.41	< 0.001
R20C15	21.15	-11.19	0.02	33.04	-4.26	0.01	34.03	-3.34	0.01	3.65	-14.72	0.06	3.62	-10.62	< 0.001	3.62	-10.62	< 0.001
R25C15	18.54	-22.12	< 0.001	31.70	-8.15	< 0.001	33.65	-4.43	< 0.001	3.59	-16.12	0.03	3.56	-12.10	0.06	3.56	-12.10	0.06

3.3. Effect of RHA and CHA on Split Tensile and Flexural Strength. As shown in Figure 3, for RHA, the STS improves significantly at lower replacement levels (5%–10%) with values reaching 4.25 MPa at 10% RHA, compared to the control mix which has a tensile strength of 4.05 MPa. This is in line with the previous research, where the fine particles of RHA were found to contribute to the densification and refinement of the concrete microstructure, leading to enhanced tensile strength [11, 23]. Khan and Haq [25] similarly reported an increase in STS for both binary and ternary blends containing RHA, MK, and SF, attributing this to the improved microstructure and bonding within the matrix. Swaminathan et al. [18] also observed a parallel improvement in tensile strength in HPC mixes with RHA and MK, consistent with their compressive strength outcomes. However, when the RHA content exceeds 15%, a decline in tensile strength is observed, with 25% RHA showing a strength of 4.01 MPa. This decline is likely due to excessive water absorption by RHA particles, reducing workability and hydration efficiency, thus leading to lower strength. For CHA, the effect on STS is more modest. At 5% CHA, the tensile strength increases slightly to 4.29 MPa. However, when CHA is used at higher levels (10%–15%), the tensile strength begins to decrease significantly. This is likely because CHA has lower pozzolanic reactivity compared to RHA, and its coarser particle size does not contribute as effectively to enhancing the microstructure of the concrete.

When both RHA and CHA are combined, the results demonstrate a synergistic effect at moderate replacement levels. The optimal combination of 10% RHA and 5% CHA (R10C5) produces the highest STS of 4.35 MPa with an 7.41% increment from the control, as shown in Table 4. This combination effectively balances RHA's pozzolanic reactivity and CHA's coarser particles, which improves workability and enhances overall strength. However, when higher levels of both materials are used, such as 25% RHA and 15% CHA (R25C15), the tensile strength significantly decreases to 3.56 MPa, which represents a decrease of 12.10% from the control. This is likely due to the negative impacts of both materials when overused, which reduces hydration efficiency and bonding within the concrete matrix. These findings align with studies by Demissew et al. [15] and Asfaw et al. [14], which highlighted that CHA could enhance the STS of concrete at moderate replacement levels, though excessive CHA reduces its effectiveness. Tarekegn et al. [24] also noted a general decline in STS with increasing hybrid ash (CHA and sugarcane bagasse) content, although mixes with up to 10% replacement still surpassed the target strength. Similarly, research by Zareei et al. [8] and Raman et al. [23] on RHA confirmed that its fine particles contribute to improved interparticle bonding and enhanced concrete properties, although excessive RHA reduces workability and hydration.

The flexural strength results presented in Figure 4 highlight how varying percentages of RHA and CHA as partial replacements for OPC impact the concrete's ability to withstand bending stresses. The general trend reveals that the flexural strength varies significantly with different replacement levels, with some combinations leading to an

increase in strength, while others result in a decrease. The highest flexural strength values are observed at moderate percentages of RHA and CHA. For example, the combination of 10% RHA and 5% CHA (R10C5) shows the highest flexural strength of 4.77 MPa with a 11.45% increment from the control concrete (R0C0) value of 4.28 MPa, followed by 15% RHA and 5% CHA (R15C5), which results in 4.74 MPa with a 10.75% increment from the control concrete (Table 4). These findings align with Tarekegn et al. [24], who reported that moderate amounts of CHA and sugarcane bagasse improve flexural strength by forming additional C–S–H gel, enhancing microstructure, and bonding, and Siddika et al. [26], who found that moderate amounts of RHA contribute to flexural strength by filling voids, improving interparticle bonding, and hydration.

As the percentage of RHA and CHA increases, a decrease in flexural strength is noticeable. For instance, 20% RHA and 15% CHA (R20C15) result in a flexural strength of 3.65 MPa, and 25% RHA and 15% CHA (R25C15) see a further drop to 3.59 MPa. This reduction can be attributed to the excessive presence of both RHA and CHA, which can interfere with the hydration process and create a stiffer, less workable mix. While RHA contributes to the pozzolanic reaction, excessive amounts can absorb too much water, making the mixture difficult to handle and reducing its bonding capability, which ultimately lowers its flexural strength. CHA, being coarser and less reactive than RHA, does not fully contribute to strength when used in high amounts, further limiting the concrete's performance. This trend corresponds with the findings of Demissew et al. [15] and Akeke et al. [27], who found that high replacement levels of CHA and RHA can reduce concrete's performance due to the reduced availability of water for hydration and the decreased effectiveness of the pozzolanic reactions at higher dosages.

RHA, with its fine particles and high silica content, significantly contributes to the strength of concrete through its pozzolanic properties; however, excessive use can lead to reduced workability and hydration efficiency. CHA, although less reactive, helps maintain workability and provides some benefits when used in moderation. As the percentages of RHA and CHA rise, their individual contributions to strength become less effective, and the resulting concrete exhibits lower flexural strength. This observation aligns with the studies of Gedefaw et al. [28] and Akeke et al. [27], who noted that while moderate use of RHA and CHA can improve concrete performance, excessive incorporation of these materials diminishes the concrete's ability to form a dense, homogeneous microstructure, leading to a reduction in flexural strength.

3.4. Effect of RHA and CHA on Water Absorption. In this study, the effect of replacing cement with varying amounts of RHA and CHA on the water absorption of C-25 concrete was investigated over 7, 28, and 56 days, and the results are presented in Figure 5. The results demonstrate that both RHA and CHA influence the water absorption of concrete, with the combination of the two materials generally leading to a reduction in water absorption over time. For the mix

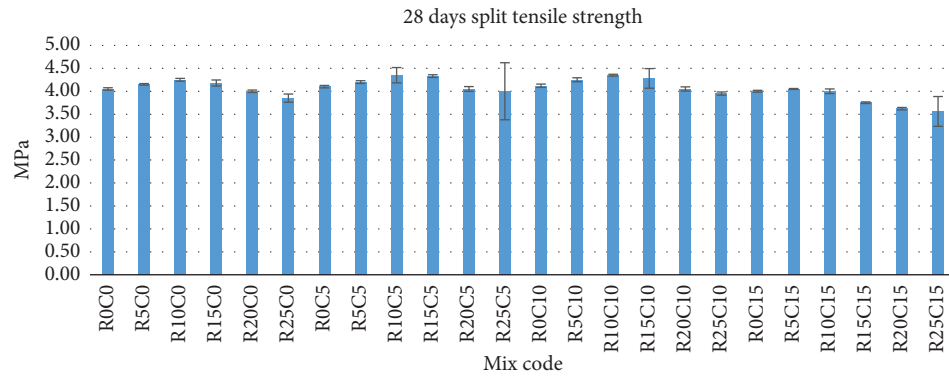


FIGURE 3: Split tensile strength of concrete with varying percentages of RHA and CHA.

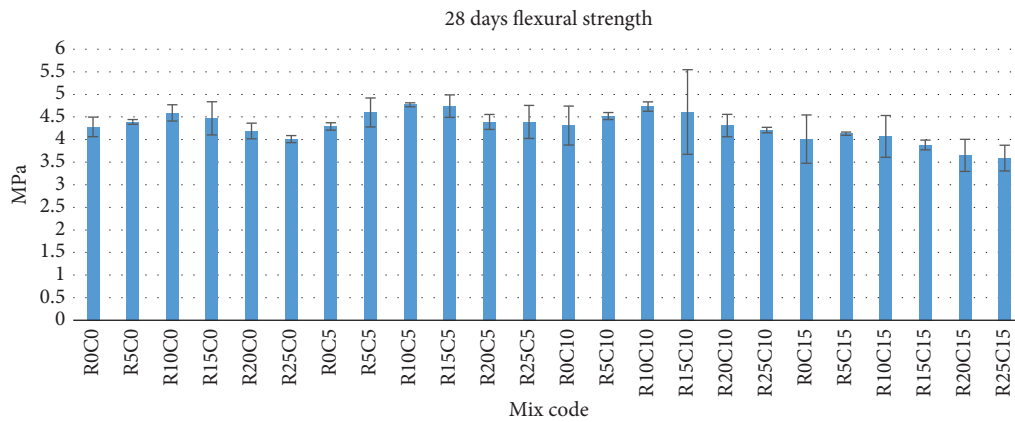


FIGURE 4: Flexural strength of concrete with varying percentages of RHA and CHA.

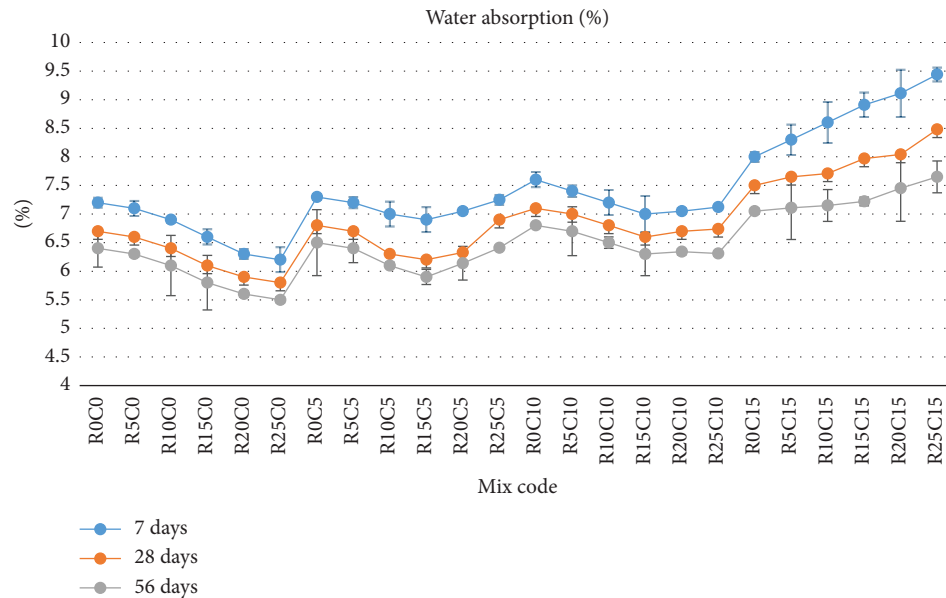


FIGURE 5: Water absorption % for 7, 28, and 56 days.

R0C0, containing no RHA or CHA, the water absorption is 7.20% at 7 days and decreases slightly to 6.40% at 56 days. This decrease over time is expected as the hydration products continue to develop and mature, leading to a denser

microstructure. Previous studies have shown that as hydration progresses, the formation of C–S–H gel reduces porosity and enhances the water resistance of concrete [23]. Additionally, RHA has been reported to reduce permeability

[29, 30]. Khan and Haq [25] reported a significant reduction in water absorption with mix (10% RHA + 10% SF) showing the lowest values, indicating enhanced resistance to moisture ingress. Similarly, Kanthe et al. [17] observed that ternary blends with RHA and FA exhibited lower water absorption, attributing these improvements to a denser microstructure resulting from finer particle filling and additional C–S–H gel formation.

However, when RHA is added, such as in the R25C0 mix with 25% RHA, the water absorption at 7 days is 6.2%, which is slightly lower than that of the R0C0 mix, and it decreases to 5.5% at 56 days. The presence of RHA in the mix introduces pozzolanic reactions, where the silica in RHA reacts with calcium hydroxide (CH) produced during cement hydration, forming additional calcium silicate hydrate (C–S–H) gel. This increases the density of the concrete, filling voids and reducing its permeability. Several studies have shown that RHA's silica reacts with CH to form additional C–S–H, which significantly improves the microstructure, leading to better durability [23, 31]. However, at higher RHA contents, excessive replacement can lead to reduced strength and durability due to a less dense matrix [6, 20].

The addition of CHA in the mix, especially when combined with RHA, leads to a more significant improvement in water absorption. In the mix R15C5, which contains 15% RHA and 5% CHA, the water absorption at 7 days is 6.90%, which decreases to 6.20% at 28 days and 5.90% at 56 days. This marked reduction in water absorption can be attributed to the synergistic pozzolanic effects of both RHA and CHA. CHA, like RHA, contains high levels of silica and other reactive compounds that contribute to the formation of C–S–H gel. Studies by Reta and Mahto [16] have demonstrated that CHA's pozzolanic activity improves the microstructure and durability of concrete, particularly in terms of reducing pore connectivity. The inclusion of CHA improves the microstructure by filling voids and reducing the permeability of the concrete.

In comparison, the mix R25C15, containing 25% RHA and 15% CHA, shows a water absorption of 9.44% at 7 days, which decreases to 7.65% at 56 days. Although this mix benefits from the pozzolanic properties of both materials, the high percentage of RHA may cause an increase in the water demand of the mix, which can lead to a less efficient hydration process. Additionally, higher levels of both RHA and CHA may lead to an excess of fine particles, which could affect the performance of the concrete. Previous research suggests that excessive RHA content (above 15%) can lead to reduced strength and durability due to overreplacement of cement, which weakens the matrix and increases susceptibility to cracking and water absorption [6, 20].

3.5. Weight and Strength Loss of Concrete Exposed to 5% Sulfuric Acid for 28 Days. In Figure 6, the current study presents the experimental results on the compressive strength loss of concrete samples containing varying percentages of RHA and CHA after exposure to 5% sulfuric acid for 28 days. The control mix (R0C0), which did not contain

any SCMs, experienced a strength loss of 23.01%, which is typical for concrete exposed to acidic conditions. This degradation is primarily due to the reaction between sulfuric acid and calcium hydroxide in the cement matrix, leading to the formation of gypsum and weakening the structure. Previous studies have highlighted that incorporating RHA enhances the concrete's resistance to acid attack due to the reactive silica content, which reacts with calcium hydroxide to form additional C–S–H gel [23]. The results of this study align with these findings, demonstrating a reduced strength loss when RHA and CHA are included in the concrete mix, suggesting that both materials enhance the microstructure and reduce porosity, making the concrete more resistant to sulfuric acid degradation.

As observed in Figure 6, the mix containing 15% RHA and 5% CHA (R15C5) exhibited lower strength losses, showing the lowest strength loss at 17.01%. The performance was further influenced by the dosage of the supporting information, with lower dosages of RHA (5%–15%) and CHA (5%–10%) leading to better acid resistance. Similar trends have been observed in previous studies, where moderate RHA content improved resistance to environmental degradation due to a denser microstructure [29, 31]. However, higher dosages of RHA (20%–25%) resulted in higher strength losses, which corresponds with findings from Bixapathi and Saravanan [20] and Habeeb and Mahmud [6], who noted that excessive RHA content could weaken the concrete's matrix and increase susceptibility to degradation.

Notably, the mix with 15% RHA and 5% CHA (R15C5) achieved the lowest strength loss at 17.01%, demonstrating the potential of combining these two materials to enhance acid resistance. This is consistent with Reta and Mahto [16], who found that the pozzolanic activity of CHA contributed to the formation of additional C–S–H gel, resulting in a denser microstructure and reduced porosity. This synergy between RHA and CHA in the present study mirrors similar findings, where the combination of these two materials led to improved performance in aggressive environments, suggesting a beneficial interaction between the materials.

The results of this study in Figure 7 clearly show the beneficial effects of incorporating RHA and CHA as SCMs in concrete exposed to 5% sulfuric acid for 28 days in terms of weight loss resistance. The concrete mix without any SCMs (R0C0) experienced the highest weight loss (5.03%) due to the direct reaction between sulfuric acid and the calcium compounds in the concrete, leading to the formation of gypsum and the deterioration of the concrete structure. This weight loss is expected as sulfuric acid typically reacts with calcium hydroxide and calcium silicate hydrate in concrete, weakening the matrix, a result that aligns with the observations of Raman et al. [23].

When RHA was introduced, the weight loss gradually decreased as the percentage of RHA increased from 5% to 25%. This can be attributed to the high silica content in RHA, which reacts with calcium hydroxide in concrete to form additional calcium silicate hydrate (C–S–H) gel, improving the concrete's microstructure and reducing permeability. This finding is supported by Ganesan et al. [31] and Bui et al.

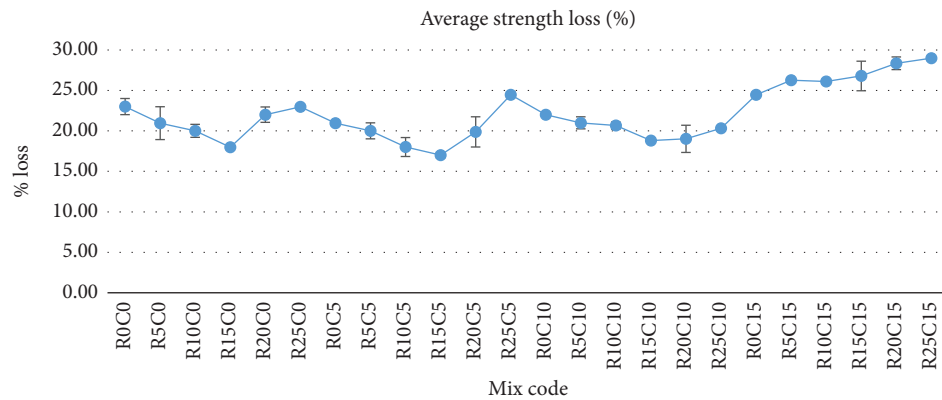


FIGURE 6: Strength loss % after 28 days of 5% sulfuric acid exposure.

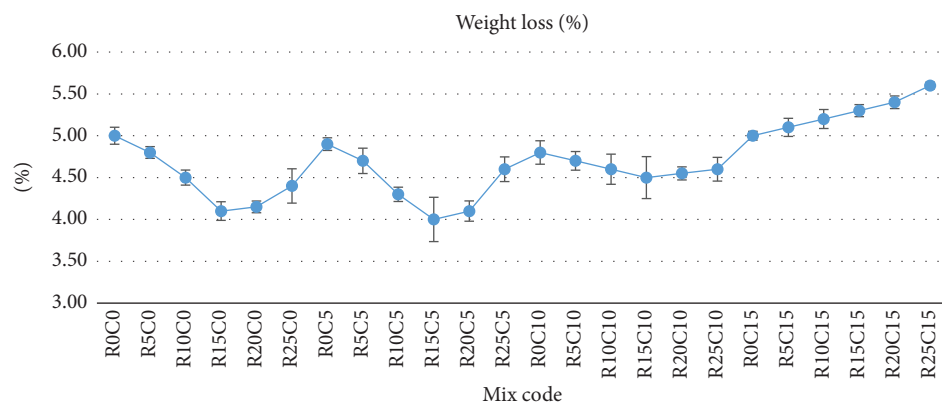


FIGURE 7: Weight loss % after 28 days of 5% sulfuric acid exposure.

[29], who also reported that RHA improved the pore structure, making the concrete denser and more durable. Sulfate resistance was a focal point in the study by Chin-daprasirt et al. [13], where mortar bars containing RHA and FA showed significantly reduced expansion in a sodium sulfate solution. RHA was found to be more effective than FA in mitigating sulfate attack, likely due to a more substantial reduction in calcium hydroxide and a lower C/S ratio in the C-S-H gel.

Interestingly, when both RHA and CHA were combined in the mix, there was a synergistic effect, leading to an even lower weight loss compared to the individual use of either material. For instance, the mix containing 15% RHA and 5% CHA (R15C5) showed a weight loss of 3.73%, which was slightly better than the individual mixes of 15% RHA and 5% CHA, indicating that the combination of both materials can enhance the concrete's acid resistance further. This indicates the contribution of CHA, as supported by Reta and Mahto [16], who found that using CHA as a partial cement replacement improved concrete durability. This was attributed to CHA's pozzolanic activity, which formed additional C-S-H gel, resulting in a denser microstructure with reduced pore connectivity and enhanced resistance to degradation. Despite these positive results, the study also shows that at the highest dosages of RHA and CHA, the weight loss

starts to increase slightly. At higher dosages (20%–25%), the reduction in weight loss becomes less pronounced, as excessive RHA content can weaken the binding capacity of the mix, leading to slightly reduced initial compressive strength and a minor increase in weight loss, as observed by Bixapathi and Saravanan [20] and Habeeb and Mahmud [6]. Similarly, the inclusion of CHA in the concrete mixes resulted in reduced weight loss compared to the control mix.

Like RHA, CHA's silica content plays a key role in enhancing the concrete's acid resistance by improving the formation of C-S-H gel, which densifies the concrete's matrix. This was confirmed by Reta and Mahto [16], who found that CHA contributed to the formation of additional C-S-H gel, leading to reduced pore connectivity and better resistance to acid attack. However, CHA's reactivity is slightly lower than that of RHA, and at higher dosages (10%–15%), the benefits of CHA begin to level off, and the weight loss values are closer to those of the RHA mixes. Therefore, there is a need to balance the use of RHA and CHA in the concrete mix to achieve both strength and durability, particularly in acidic environments. This balance is consistent with findings from Bixapathi and Saravanan [20] and Habeeb and Mahmud [6], who indicated that excessive amounts of supporting information can compromise the concrete's structural integrity.

3.6. Microstructural Analysis Using SEM. The microstructure of the concrete samples was examined using SEM to evaluate the effects of incorporating RHA and CHA on hydration products and matrix densification. The SEM micrographs of the control mix and two experimental mixes are presented in Figures 8(a), 8(b), 8(c).

Figure 8(a) shows the microstructure of the control concrete composed of OPC without any SCMs. The image reveals a porous and relatively loose matrix structure, with prominent crystalline formations indicative of portlandite ($\text{Ca}(\text{OH})_2$) and unhydrated cement particles. The presence of these larger, unreacted phases suggests limited secondary hydration and suboptimal pozzolanic activity. The microstructure lacks the dense, amorphous calcium silicate hydrate (C–S–H) gel that is typically responsible for mechanical strength and durability. The high visibility of $\text{Ca}(\text{OH})_2$ also correlates with increased vulnerability to chemical degradation, such as sulfate or acid attack, which limits the long-term performance of the control mix. In contrast, the SEM image of the concrete with 10% RHA and 5% CHA replacement, shown in Figure 8(b), exhibits a significantly more compact and homogeneous microstructure. The matrix is characterized by reduced porosity and abundant amorphous gel-like phases that are indicative of extensive pozzolanic reaction products. The high silica content of RHA (80.72%) facilitates the consumption of $\text{Ca}(\text{OH})_2$ to form additional C–S–H, while the alumina (7.40%) and iron oxide (3.50%) from CHA contribute to the formation of calcium aluminate hydrate (C–A–H) and calcium ferrite hydrate (C–F–H), respectively. The synergistic interaction between RHA and CHA results in a total reactive oxide content ($\text{SiO}_2 + \text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3$) of 76.62%, supporting high pozzolanic reactivity as per ASTM C618 criteria. These reactions contribute to the improved compressive strength observed at 56 days, as well as enhancements in tensile and flexural properties. Moreover, the reduced visibility of $\text{Ca}(\text{OH})_2$ crystals in Figure 8(b) confirms its consumption and effective transformation into strength-contributing hydrates. These observations are consistent with previous findings. For instance, Khan and Haq [25] demonstrated that ternary blends containing RHA, SF, and MK formed additional C–S–H gel and reduced porosity, resulting in improved long-term durability. Similarly, Kanthe et al. [17] showed that binary and ternary mixes with RHA and FA exhibited denser microstructures due to enhanced particle packing and pozzolanic reactivity.

Figure 8(c) presents the SEM micrograph of the concrete sample with 15% RHA and 5% CHA (R15C5). This mix shows the densest microstructure among all the samples, characterized by a highly refined and tightly packed hydration product matrix with minimal visible voids or cracks. The increase in RHA content leads to a higher degree of pozzolanic reaction and further reduction in free $\text{Ca}(\text{OH})_2$, resulting in an extensively developed C–S–H network. The combined effect of the increased reactive silica from RHA and the supplementary oxides from CHA promotes enhanced matrix densification, thereby significantly improving durability properties such as water absorption resistance and sulfuric acid resistance. Chindaprasirt et al. [13] similarly

observed that RHA effectively reduced sulfate attack by lowering calcium hydroxide content and altering the C/S ratio in the C–S–H gel, creating a more chemically stable matrix with reduced ettringite formation. However, despite the improved microstructure, the mechanical strength does not continue to increase beyond the 10% RHA level. This is attributed to the dilution effect, wherein excessive replacement reduces the quantity of primary cementitious material, coupled with increased water demand due to RHA's high surface area. The relatively coarser CHA particles may also interfere with uniform compaction at higher replacement levels.

The SEM analysis supports the mechanical and durability trends observed in this study. The optimized blend of 10% RHA and 5% CHA achieves a favorable balance between hydration, pore refinement, and reactive oxide composition, while the 15% RHA and 5% CHA mix further enhances durability through superior matrix densification. The optimal mechanical performance observed at 10% RHA and 5% CHA is primarily due to the high silica (SiO_2) content in RHA, which is 80.72%, and its finer particle size that enhances pozzolanic activity, promotes better particle packing, and contributes to the formation of additional calcium silicate hydrate (C–S–H), thereby increasing compressive, tensile, and flexural strengths. CHA, though coarser in particle size and lower in silica at 59.85%, compensates with higher amounts of alumina (Al_2O_3 at 7.40%) and iron oxide (Fe_2O_3 at 3.50%), which further contribute to strength development through secondary pozzolanic reactions. The combination of 10% RHA and 5% CHA results in an effective total of reactive oxides ($\text{SiO}_2 + \text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3$) of approximately 76.62%, creating a balance between reactivity and workability that supports optimal hydration and matrix densification. However, durability is observed to be highest at a slightly higher replacement level of 15% RHA and 5% CHA due to the increased pozzolanic consumption of calcium hydroxide, resulting in a denser microstructure with lower porosity and permeability. The higher SiO_2 contribution from increased RHA content enhances the durability characteristics by further refining the pore structure and reducing water absorption and acid ingress. Despite this, mechanical performance does not continue to improve beyond the 10% RHA level due to the dilution effect—where excessive replacement reduces the cementitious material—and increased water demand from the highly reactive and fine RHA. Moreover, CHA's coarser particles can interfere with compaction and homogeneity at higher dosages. Therefore, the reversal in optimal replacement levels for mechanical strength and durability is a result of the interplay between oxide composition—particularly SiO_2 , Al_2O_3 , and Fe_2O_3 —and the physical characteristics of the ash materials, emphasizing the need for a tailored approach to mix design in concrete using agrowaste materials.

3.7. Implications of Statistical Results on Concrete Performance for R10C5 vs. R15C5. The two-sample *t* test (independent *t* test) was used to assess whether there were statistically

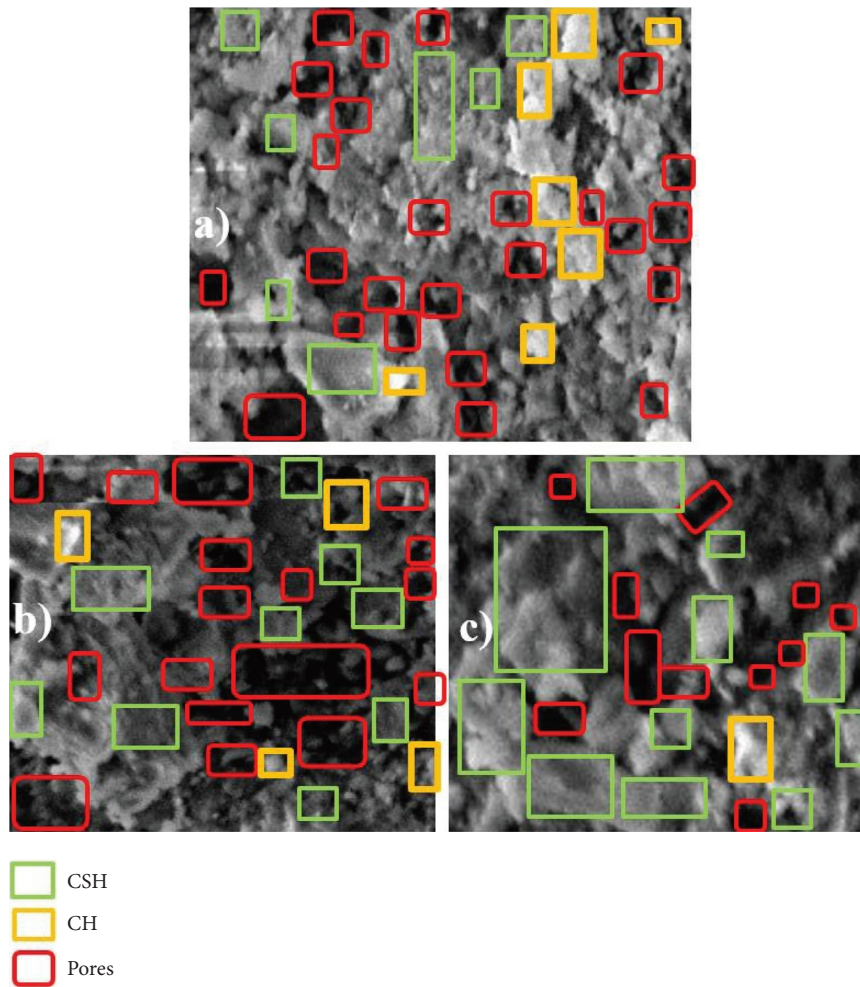


FIGURE 8: SEM images at 6000 magnification: (a) control, (b) R10C5, and (c) R15C5 mixes, using 28-day specimens.

significant differences between the means of the two groups (R10C5 vs. R15C5) for each of the properties listed in the study. Since there was a possibility of unequal variances between the groups, we applied Welch's t test, which adjusts for this variance inequality. As shown in Table 5, the results reveal no significant differences between the two mixes (R10C5 and R15C5) for all the properties tested, as all p values are greater than the 0.05 significance threshold. This suggests that, based on the statistical analysis, the two mixes exhibit similar performance in terms of compressive strength, tensile strength, flexural strength, water absorption, water loss, and sulfate resistance. However, microstructural observations present some differences. The R15C5 mix exhibits a denser and more refined microstructure compared to R10C5, which is beneficial for durability. Despite this, the mechanical strength of R15C5 does not significantly outperform R10C5, likely due to the dilution effect from excessive replacement levels and increased water demand from the finer RHA particles. Therefore, the results indicate that while R10C5 offers an optimal balance between

TABLE 5: Results of statistical tests (p values).

Test	p value
7-day compressive strength (MPa)	0.118
28-day compressive strength (MPa)	0.914
56-day compressive strength (MPa)	0.883
28-day flexural strength (MPa)	0.855
28-day split tensile strength (MPa)	0.856
Water absorption 7 days (%)	0.604
Water absorption 28 days (%)	0.865
Water absorption e 56 days (%)	0.106
Weight loss 28 days (%)	0.180
Strength loss 28 days (%)	0.274

mechanical strength and durability, R15C5 achieves superior durability, although at a slight compromise in mechanical strength. Depending on the project's specific needs, R10C5 may be more suitable for applications focusing on mechanical performance, while R15C5 could be better for projects where durability is the primary concern.

4. Conclusions

In conclusion, this experimental investigation into the effects of partial cement replacement with RHA and CHA has provided significant insights into how these agrowaste materials influence concrete properties. The inclusion of both RHA and CHA was shown to significantly affect workability, mechanical properties, and durability of concrete. RHA, due to its finer particles, increased surface area, and higher water absorption, had a more pronounced impact on reducing workability compared to CHA. This result can be attributed to RHA's ability to absorb more water, thus decreasing the ease with which the concrete mix can be worked. In contrast, CHA had a relatively smaller effect on workability, likely due to its coarser particles and lower water absorption characteristics.

The study also revealed that moderate dosages of RHA and CHA contributed positively to compressive strength. The optimal mix of 10% RHA and 5% CHA led to substantial improvements in compressive strength, with increases of 5.04% and 7.87% at 28 and 56 days, respectively. The combination enhanced the concrete's microstructure and interparticle bonding, leading to a denser and stronger material. However, at higher replacement levels, particularly above 15% for CHA and 25% for RHA, the strength decreased. This negative effect is likely due to the disruption of cement hydration processes, where excessive additions of RHA and CHA may limit the available cementitious material for proper hydration, thus impairing strength development.

In terms of flexural strength and STS, the optimal combination of 10% RHA and 5% CHA yielded the highest strengths of 4.77 MPa and 4.35 MPa, respectively. These values represent an increase of 11.45% in flexural strength and a 7.41% improvement in STS compared to the control concrete. These improvements can be attributed to the same microstructural enhancements seen in compressive strength, which positively influenced the concrete's ability to resist tensile and bending forces. However, as the dosage of RHA and CHA exceeded the optimal levels, strength in these areas declined. This can be explained by the adverse effects of excessive water demand, poor workability, and the interference of high ash content with the hydration process, all of which hinder the material's performance.

Durability was notably enhanced by the inclusion of RHA and CHA, especially in terms of water resistance. The mix containing 15% RHA and 5% CHA (R15C5) emerged as the most effective in reducing water absorption. This improvement can be attributed to the synergistic pozzolanic reactions between RHA and CHA, which promoted the formation of additional calcium silicate hydrate (C-S-H) gel, leading to a denser and less permeable microstructure. Such a structure is crucial for enhancing the concrete's long-term durability and resistance to environmental degradation. Further, the optimal performance in sulfuric acid resistance was observed with the same R15C5 mix, which exhibited the lowest strength and weight loss. This reinforces the idea that the combined use of RHA and CHA results in a synergistic effect that improves the durability of concrete in aggressive environments. However, excessive use of RHA

(above 20%) or CHA (above 10%) was found to compromise performance, likely due to the negative effects on both hydration and workability. The SEM analysis confirms that incorporating RHA and CHA into concrete positively influences microstructural development, hydration behavior, and durability. The mix with 10% RHA and 5% CHA (R10C5) achieves an optimal balance, enhancing both mechanical strength and microstructural refinement due to high pozzolanic reactivity and effective oxide composition. Although the R15C5 mix has a denser matrix and improved durability, it does not significantly outperform R10C5 in mechanical properties due to the dilution effect and increased water demand. Statistical analysis confirms similar performance between the two mixes, suggesting that R10C5 is ideal for strength-focused applications, while R15C5 is better for environments needing enhanced durability, despite no significant difference in performance.

Further research should focus on using superplasticizers to offset the reduced workability caused by RHA and on optimizing water-to-cement ratios to maintain performance. Advanced rheological testing and techniques like TG/DTA or isothermal calorimetry can provide deeper insight into flow behavior and hydration kinetics. Additionally, long-term durability assessments—such as resistance to chloride ingress, carbonation, and freeze-thaw cycles—along with field trials and lifecycle assessments, are crucial for refining mix designs and promoting the sustainable use of agrowaste materials in concrete.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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